

Hazem Mohammad Salah

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SUMMARY

AI Engineer and Data Scientist focused on Generative AI and Arabic NLP. Built and deployed a production Arabic RAG chatbot over 50,000+ fatwa records and published the dataset on Kaggle. Completed NLP and Data Science internships and NVIDIA DLI training in LLM applications and RAG. Strong in Python, SQL, embeddings-based retrieval, and machine learning, working in Arabic and English. Open to junior AI / NLP / GenAI / Data Science roles.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, Java, C++, C#, HTML, CSS

GenAI, LLMs & NLP: RAG, LangChain, Prompt Engineering, Llama 3.3 (Groq), Gemma (Ollama), Embeddings (Multilingual E5, CAMEL-Lab BERT, BGE-M3), Semantic Search, Intent Routing, spaCy, NLTK, TF-IDF, Sentiment Analysis, NER, Text Classification

Vector DBs & Backend: ChromaDB, FastAPI, REST APIs, Webhooks, WhatsApp Cloud API, Docker, Streamlit, Git, GitHub

ML & Data Science: Scikit-learn, PyTorch, TensorFlow, KNN, Feature Engineering, Pandas, NumPy, Statistics, EDA, Data Cleaning

Data Engineering & BI: ETL Pipelines, Web Scraping, SQL Server, SQLite, Power BI (DAX), Matplotlib, Seaborn

Currently Learning: Agentic AI systems, LLM/RAG evaluation (Ragas, DeepEval)

EXPERIENCE

Agentic AI & Generative AI System Developer Trainee · DEPI (Digital Egypt Pioneers Initiative) · Remote · Ongoing

- Training program covering LLM architecture and debugging, API integration, advanced prompting, Hugging Face models, RAG and memory systems, multi-agent systems, and deployment and monitoring.

NLP Intern · Elevvo · Remote · March 2026 (1 month)

- Completed four hands-on NLP tasks: sentiment analysis, news category classification, fake news detection, and named entity recognition, using TF-IDF, Scikit-learn, NLTK, and spaCy.

Data Scientist Intern · DEPI (Digital Egypt Pioneers Initiative) · Remote · June 2025 – Dec 2025

- Developed and deployed predictive machine learning models integrated into analytical pipelines to support business decision-making.
- Built ETL workflows to clean, preprocess, and transform large-scale datasets, ensuring high data consistency.
- Designed interactive Power BI dashboards and automated reporting, and communicated insights to stakeholders.

PROJECTS

Online Efteely: Production Arabic RAG Chatbot (live on Hugging Face Spaces)

- Retrieval pipeline with ChromaDB over 50,000+ fatwa records using multilingual-e5 embeddings; LangChain chain with Llama 3.3 70B (Groq), answers grounded only in retrieved fatwas.
- Conversation-aware retrieval (last 6 turns), LLM-based intent detection separating chitchat from religious questions, and source links to the original fatwas.

Efteely Offline + 50K Arabic Fatwa Dataset (Kaggle)

- Scraped and preprocessed 51,000+ fatwas from three sources and published the dataset on Kaggle.
- Offline RAG assistant: CAMEL-Lab Arabic BERT embeddings (768-d) with cosine-similarity top-5 retrieval and a local Gemma 3 4B LLM via Ollama.

WhatsApp RAG Customer-Service Bot (personal project)

- RAG bot over a product knowledge base (ChromaDB, multilingual-e5, Llama 3.3 via Groq) connected to the WhatsApp Cloud API through FastAPI webhooks and Docker; tested end-to-end, preparing cloud deployment.

MAZAG Movies: KNN Recommendation System · Automated Crypto Tracker

- MAZAG: Pandas and Scikit-learn KNN model with feature engineering and scaling on a 7,000+ movie dataset; web app built with AI-assisted development.
- Crypto Tracker: Python tool pulling live prices through REST APIs into SQLite, with logging and error handling.

TRAINING & CERTIFICATIONS

- NVIDIA DLI Generative AI Training Program (ITI, 35 hours, Jan–Feb 2026): Prompt Engineering for LLM applications (17h) and Augmenting LLMs with RAG (9h).
- NVIDIA Academy: AI for All, From Basics to GenAI Practice (Jan 2026) · DEPI Data Scientist track certificate, MCIT (2025) · Elevvo NLP Internship (Mar 2026) · Oracle AI for You badge (100%).

EDUCATION

Minya National University, Egypt · B.Sc. in Data Science · 2023 – 2027 (Expected)

Coursework: Machine Learning, NLP, Artificial Intelligence, Data Mining, Information Retrieval and Web Search, Statistics, Software Development, IoT, Database Management, Image Processing

LANGUAGES

Arabic (Native), English